

arc_temp_humid — outstanding work

Written 6 September 2026, at the end of the rebrand and ARC UK build-out.

Everything described below is **not yet done**. What *is* done is recorded in `CHANGELOG.md` ; the architecture is in `CLAUDE.md` .

1. Rename the GitHub repository

`actionresearchprojects/arc_tz_temp_humid` → `arc_temp_humid` .

The local folder, the git remote and every in-repo reference are already updated. The rename itself has not been done, because it is outward-facing.

What the rename will and will not break

GitHub permanently redirects the old repository URL, for both web and git, so clones, fetches, existing PATs and repository secrets all keep working. Actions and their secrets survive a rename untouched.

The damage is elsewhere, and it is silent. The live dashboard reaches the public site through this chain:

```
actionresearchprojects.net/graphs/arc-tz-temp-humid/  
└─ iframe → ../../embedded/arc_tz_temp_humid/index.html  
      ↑ written by sync-embedded.yml on the main site,  
      into a folder named after the source repo
```

`sync-embedded.yml` derives the destination folder straight from the `source_repo` value in the dispatch payload. That payload now says `arc_temp_humid` (updated in both workflows in this repo), so after the rename the next sync will:

1. clone `github.com/actionresearchprojects/arc_temp_humid.git` — fine;
2. write into a **new** folder, `embedded/arc_temp_humid/` ;

3. leave `embedded/arc_tz_temp_humid/` in place, frozen at whatever it held on the day of the rename.

The public page still iframes the **old** folder. So nothing 404s and nothing errors — the dashboard simply stops updating, showing stale data indefinitely. That is the worst failure mode available, because nobody notices.

The main-site changes that must accompany the rename

All in `actionresearchprojects/actionresearchprojects.github.io`:

#	Change	Why
1	<code>graphs/arc-tz-temp-humid/</code> → <code>graphs/arc-temp-humid/</code>	Public URL drops "tz" to match the rebrand
2	In that page, iframe <code>src</code> → <code>../../embedded/arc_temp_humid/index.html</code>	Otherwise it keeps pointing at the frozen folder
3	Delete <code>embedded/arc_tz_temp_humid/</code> once the first new sync has run	Dead copy, ~16 MB
4	<code>explainers/arc-tz-temp-humid/</code> → <code>explainers/arc-temp-humid/</code> , then flip the header info-icon link in <code>build.py</code> to match	The dashboard's info icon deliberately still points at the old slug, so nothing 404s in the meantime
5	Redirect the old <code>/graphs/arc-tz-temp-humid/</code> URL to the new one	Existing links and any published references
6	Update remaining mentions: <code>README.md</code> , <code>explainers/data-flow/viewer.html</code> , <code>explainers/arc-tz-weather/viewer.html</code>	Stale cross-links

Item 4 is deliberately two-sided. The dashboard's header info icon was left pointing at the old explainer slug, because that path still works and switching it early would have shipped a live 404. Rename the explainer folder first, then change the one link in `build.py` (it is commented to say so) and rebuild.

Recommended order

1. Make the main-site changes (1, 2, 4, 5, 6) **first**, pointing at the new folder names before they exist. The site briefly serves the old embedded copy from a new path, which is harmless.
2. Rename the repository.
3. Trigger a sync (push anything, or run `update-dashboard-data.yml` manually) and confirm `embedded/arc_temp_humid/` appears.
4. Delete `embedded/arc_tz_temp_humid/` (item 3).

Also check that the Pages URL change is acceptable: this repo publishes to `actionresearchprojects.net/arc_tz_temp_humid/`, which becomes `actionresearchprojects.net/arc_temp_humid/`.

2. Automate the ARC UK Omnisense fetch

The UK export is currently added **by hand** to

`data/omnisense_uk/omnisense_uk_YYYYMMDD_HHMM.csv`. The timestamp in the filename is the fetch time and is what the sidebar's freshness note reads, so keep the format.

`fetch_omnisense.py` is already parameterised by site. To switch it on:

1. Fill in the commented-out `SITES["uk"]` entry, in particular `site_nbr`.
2. Add a `--site uk` step to `.github/workflows/update-dashboard-data.yml`, alongside the existing Omnisense step and with the same `continue-on-error`.

Finding the site number

It is not in the CSV export — the export carries site *names*, not ids, and the `download_935909450.csv` filename is a download-job id, not a site number.

Log in at omnisense.com, select the site, and read `siteNbr=` from the address bar. That is exactly where the Tanzanian value came from: `152865`, visible in `fetch_omnisense.py` as `dnld_rqst.asp?siteNbr=152865`.

The site to look for is named "**Simmonds.Mills Retrofits**" — that is the `site_name` on every block of the UK export.

Credentials are probably already correct

The UK export contains a sensor labelled `House 5 Metal Roof S.E` alongside the Grove, Holywell, Chestnuts and No. 59 sensors. A single login therefore already sees both the Tanzanian and the UK material, which strongly suggests the existing `OMNISENSE_USERNAME` / `OMNISENSE_PASSWORD` secrets will work and only `site_nbr` differs.

If that turns out to be wrong, `SITES["uk"]` accepts `username_env` and `password_env` naming its own secrets; the code already falls back to the shared pair when they are absent.

One thing to check when it runs

The UK export is a single CSV covering several sites — Grove, Holywell, Chestnuts, No. 59 — of which this dashboard uses four sensors. `GROVE_SENSORS` and `HOLYWELL_SENSORS` in `build.py` filter it by sensor id, so unrelated sensors are ignored rather than plotted. If sensors are added or swapped at those sites, those two sets are what needs updating.

3. Decide the UK overheating threshold

Currently **none**: `DATASETS["grove"]["threshold"]` and the other two UK datasets are `None`, so no red band is drawn and the sidebar control is hidden rather than offering a meaningless toggle.

Tanzania keeps 32–35 °C, which is a Mkuranga heat-stress range and means nothing against UK data peaking near 29 °C.

No UK figure was assumed, because the sensible source — CIBSE TM52 / TM59 — is **per room type**. TM59 uses 26 °C for bedrooms, but Grove's sensor is a living room and Holywell's is "Internal Ambient", so a single number cannot be picked without deciding which criterion applies to which sensor.

Once decided it is one line per dataset:

```
"threshold": {"low": 26, "high": 28},    # or whatever the criterion gives
```

The label, the band on all three chart types and the sidebar control all follow from that automatically.

4. Optional: a UK Open-Meteo feed

Not needed for correctness — each UK building carries its own external ambient sensor, and that is what its adaptive comfort running mean uses. Adding Open-Meteo would give the UK datasets an external reference series and a forecast, as Tanzania has.

`fetch_openmeteo.py` has a commented-out `LOCATIONS["uk"]` entry with the shape required. Beyond filling it in and adding a `--location uk` workflow step, the UK datasets would need an Open-Meteo id in their `external_sensors` and the new CSVs loaded from the new output directory.

Note the coordinates in that template are a placeholder for Grove Cottage, Hereford. Grove and Holywell are about 150 km apart, so one location cannot serve both — Holywell Barn (Criccieth) would need its own entry.

5. Known limitation, not a task

ASHRAE 55 Section 5.4.1(a) requires that no heating is in operation. Nothing at any site records heating status, and nothing in a temperature and humidity trace reliably separates a heated room from a merely warm one.

The dashboard states this rather than inferring it: a note on the UK comfort chart, a bolded line in the sidebar applicability panel directly beneath the quoted criterion, and the tooltips explaining the method. The 10–33.5 °C running-mean gate is enforced automatically and will grey out cold-weather readings once UK winter data exists — but **a mild day with the heating on will pass every test the dashboard can apply and still sit outside the standard's scope.**

Closing this properly needs a heating-status input, not a cleverer algorithm. See `adaptivecomfort.md` §2 for the full reasoning, including why calendar-month filtering was considered and rejected.